Docker Compose Practical

1. Introduction to Docker Compose for Multi-Service Container Deployment - Deploying Nginx and Redis Containers

```
Terminal
PS C:\Users\sreev> mkdir docker-composedemo


    Directory: C:\Users\sreev


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        05-03-2026     15:52                docker-composedemo


PS C:\Users\sreev> cd docker-composedemo
RAM 0.91 GB  CPU 0.08%   Disk: 10.24 GB used (limit 1006.85 GB)
```

```
Terminal
PS C:\Users\sreev> cd docker-composedemo
PS C:\Users\sreev\docker-composedemo> code dockercompose.yml
PS C:\Users\sreev\docker-composedemo> docker compose up
no configuration file provided: not found
PS C:\Users\sreev\docker-composedemo> ren dockercompose.yml

cmdlet Rename-Item at command pipeline position 1
Supply values for the following parameters:
NewName: docker-compose.yml
PS C:\Users\sreev\docker-composedemo> docker compose up
[+] up 9/9
[+] up 12/12ls Pulled                                                    12.3s
RAM 0.91 GB  CPU 5.53%   Disk: 10.24 GB used (limit 1006.85 GB)
```

```
Terminal
✓ Image redis                        Pulled                              12.3s
✓ Network docker-composedemo_default  Created                            0.1s
✓ Container docker-composedemo-redis-1 Created                           0.7s
✓ Container docker-composedemo-web-1   Created                           0.7s

Attaching to redis-1, web-1
redis-1  | Starting Redis Server
RAM 0.91 GB  CPU 0.08%   Disk: 10.24 GB used (limit 1006.85 GB)
```

```
Terminal
redis-1  | 1:C 05 Mar 2026 10:30:04.660 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
web-1    | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration

redis-1  | 1:C 05 Mar 2026 10:30:04.660 * Redis version=8.6.1, bits=64, commit=00000000, modified=1, pid=1, just started
web-1    | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/

redis-1  | 1:C 05 Mar 2026 10:30:04.660 * Configuration loaded
RAM 0.91 GB  CPU 0.25%   Disk: 10.24 GB used (limit 1006.85 GB)
```

```
Terminal
PS C:\Users\sreev\docker-composedemo> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                      NAMES
4597b1b6c327   redis     "docker-entrypoint.s…"   2 minutes ago   Up 2 minutes   6379/tcp                                   docker-composedemo-redis-1
7098744c8f03   nginx     "/docker-entrypoint.…"   2 minutes ago   Up 2 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    docker-composedemo-web-1
PS C:\Users\sreev\docker-composedemo> docker exec -it docker-composedemo-redis-1 redis-cli
127.0.0.1:6379> ping
PONG
127.0.0.1:6379> SET city Delhi
OK
127.0.0.1:6379> Get city
"Delhi"
127.0.0.1:6379>
RAM 0.91 GB  CPU 0.17%   Disk: 10.24 GB used (limit 1006.85 GB)
```

```
Terminal

[+] stop 2/2
 ✓ Container docker-composedemo-redis-1 Stopped                    0.5s
 ✓ Container docker-composedemo-web-1   Stopped                    0.6s
PS C:\Users\sreev\docker-composedemo> type docker-compose.yml
services:
  web:
    image: nginx
    ports:
      - "8080:80"
  redis:
    image: redis
PS C:\Users\sreev\docker-composedemo>

RAM 0.88 GB  CPU 0.08%  Disk: 10.24 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

2. Using Docker Compose to Build and Run Multi-Container Applications from Dockerfiles

```
Terminal

PS C:\Users\sreev\docker-composedemo> mkdir service1


    Directory: C:\Users\sreev\docker-composedemo


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        05-03-2026     16:13                service1


PS C:\Users\sreev\docker-composedemo> mkdir service2


    Directory: C:\Users\sreev\docker-composedemo

RAM 2.52 GB  CPU 0.00%  Disk: 11.72 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

```
Terminal

PS C:\Users\sreev\docker-composedemo\service1> ren dockerfile.txt dockerfilke
PS C:\Users\sreev\docker-composedemo\service1> ren dockerfilke dockerfile
PS C:\Users\sreev\docker-composedemo\service1> ls


    Directory: C:\Users\sreev\docker-composedemo\service1


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        05-03-2026     16:15             47 app.py
-a----        05-03-2026     16:17             72 dockerfile


PS C:\Users\sreev\docker-composedemo\service1> cd
PS C:\Users\sreev\docker-composedemo\service1> cd ..
PS C:\Users\sreev\docker-composedemo> code docker-compose.yml

RAM 2.51 GB  CPU 0.00%  Disk: 11.72 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

```
Terminal

PS C:\Users\sreev\docker-composedemo\service1> cd ..
PS C:\Users\sreev\docker-composedemo> cd service2
PS C:\Users\sreev\docker-composedemo\service2> notepad dockerfile
PS C:\Users\sreev\docker-composedemo\service2> ls


    Directory: C:\Users\sreev\docker-composedemo\service2


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        05-03-2026     16:18            107 dockerfile.txt


PS C:\Users\sreev\docker-composedemo\service2> ren dockerfile.txt dockerfile

RAM 2.51 GB  CPU 0.00%  Disk: 11.72 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

```
Terminal

PS C:\Users\sreev\docker-composedemo> docker compose up --build
#1 [internal] load local bake definitions
#1 reading from stdin 1.06kB done
#1 DONE 0.0s

#2 [service1 internal] load build definition from dockerfile
#2 transferring dockerfile: 109B 0.1s done
#2 DONE 0.1s

#3 [service2 internal] load build definition from dockerfile
#3 transferring dockerfile: 144B 0.1s done
#3 DONE 0.2s

#4 [service2 internal] load metadata for docker.io/library/alpine:latest
#4 ...

RAM 2.51 GB  CPU 0.00%   Disk: 11.72 GB used (limit 1006.85 GB)
```

```
Terminal

#19 [service1] resolving provenance for metadata file
#19 DONE 0.0s
time="2026-03-05T16:23:45+05:30" level=warning msg="Found orphan containers ([docker-composedemo-redis-1 docker-composedemo-web-1]) for this project. If you removed or r
enamed this service in your compose file, you can run this command with the --remove-orphans flag to clean it up."
[+] up 4/4
 ✓ Image docker-composedemo-service2       Built                                                    61.6s
 ✓ Image docker-composedemo-service1       Built                                                    61.6s
 ✓ Container docker-composedemo-service1-1 Created                                                   0.3s
 ✓ Container docker-composedemo-service2-1 Created                                                   0.3s
Attaching to service1-1, service2-1
service2-1  | Hello from Vishnu's service2 container
service1-1  | Hello from Vishnu's Python container-1
service2-1 exited with code 0
service1-1 exited with code 0
PS C:\Users\sreev\docker-composedemo>

v View in Docker Desktop   o View Config   w Enable Watch   d Detach

RAM 2.51 GB  CPU 0.08%   Disk: 11.72 GB used (limit 1006.85 GB)
```

```
Terminal

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        05-03-2026     16:19                service1
d-----        05-03-2026     16:18                service2
-a----        05-03-2026     16:21             90 docker-compose.yml


PS C:\Users\sreev\docker-composedemo> cd service1
PS C:\Users\sreev\docker-composedemo\service1> cat dockerfile
FROM python:3.10
WORKDIR /app
COPY app.py .
CMD ["python","app.py"]
PS C:\Users\sreev\docker-composedemo\service1> cat app.py
print("Hello from Vishnu's Python container-1")
PS C:\Users\sreev\docker-composedemo\service1>

RAM 2.51 GB  CPU 0.00%   Disk: 11.72 GB used (limit 1006.85 GB)
```

```
Terminal

PS C:\Users\sreev\docker-composedemo\service2> cat dockerfile
FROM alpine
RUN echo "Installing basic utilities"
CMD ["echo","Hello from Vishnu's service2 container"]
PS C:\Users\sreev\docker-composedemo\service2> cd ..
PS C:\Users\sreev\docker-composedemo>
```

3. “Deploying a Multi-Container Flask-Redis Application using Docker Compose”

```
Terminal

PS C:\Users\sreev> mkdir composepower


    Directory: C:\Users\sreev


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        11-03-2026     13:24                composepower


PS C:\Users\sreev> cd composepower
RAM 1.23 GB  CPU 0.17%   Disk: 14.34 GB used (limit 1006.85 GB)
```

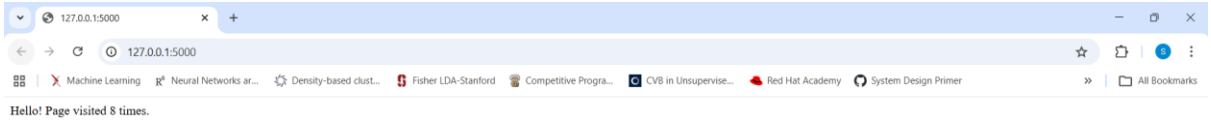
```
Terminal

PS C:\Users\sreev\composepower\app> cd ..
PS C:\Users\sreev\composepower> docker compose up --build
#1 [internal] load local bake definitions
#1 reading from stdin 521B done
#1 DONE 0.0s

#2 [internal] load build definition from dockerfile
#2 transferring dockerfile: 138B 0.1s done
#2 DONE 0.2s

#3 [internal] load metadata for docker.io/library/python:3.11
#3 ...
RAM 1.24 GB  CPU 0.08%   Disk: 14.34 GB used (limit 1006.85 GB)
```

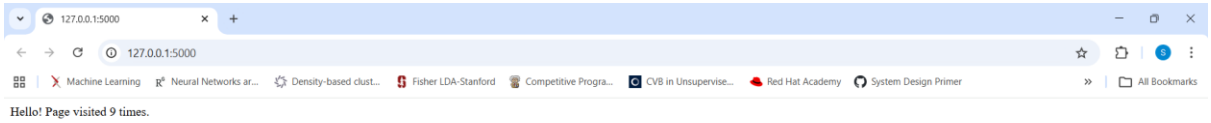
```
Terminal

#4 [auth] library/python:pull token for registry-1.docker.io
#4 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11
#3 DONE 5.9s

#5 [internal] load .dockerignore
#5 transferring context: 2B 0.0s done
#5 DONE 0.1s

#6 [1/4] FROM docker.io/library/python:3.11@sha256:33b4060c8f8268bb55f244a11adf6a1d75348f1a3a8335b858c65fe755559b29
#6 resolve docker.io/library/python:3.11@sha256:33b4060c8f8268bb55f244a11adf6a1d75348f1a3a8335b858c65fe755559b29 0.0s done
RAM 1.24 GB  CPU 0.08%   Disk: 14.34 GB used (limit 1006.85 GB)
```

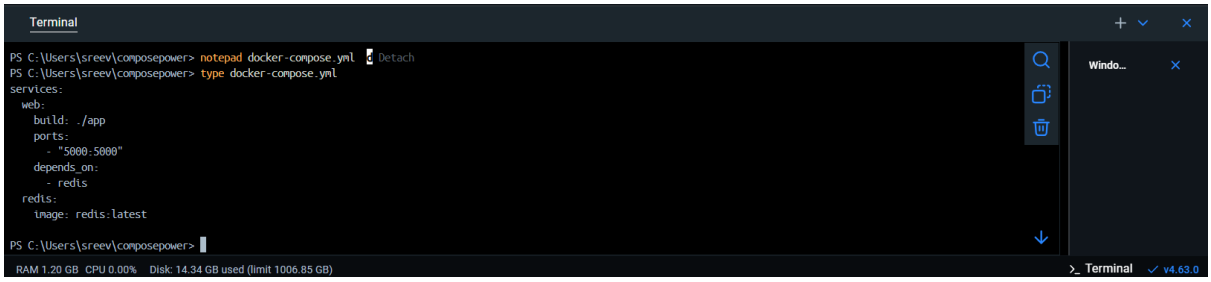
```
Terminal

#6 [1/4] FROM docker.io/library/python:3.11@sha256:33b4060c8f8268bb55f244a11adf6a1d75348f1a3a8335b858c65fe755559b29
#6 extracting sha256:d2c5d59621cc0d4d700a3681106b1bab53502b2f4ca72f3ab9254a9c08f736e0 0.0s done
#6 DONE 9.4s

#8 [2/4] WORKDIR /app
#8 DONE 0.3s

#9 [3/4] COPY app.py .
#9 DONE 0.1s

#10 [4/4] RUN pip install flask redis
#10 3.181 Collecting flask
#10 3.508   Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
RAM 1.24 GB  CPU 0.08%   Disk: 14.34 GB used (limit 1006.85 GB)
```

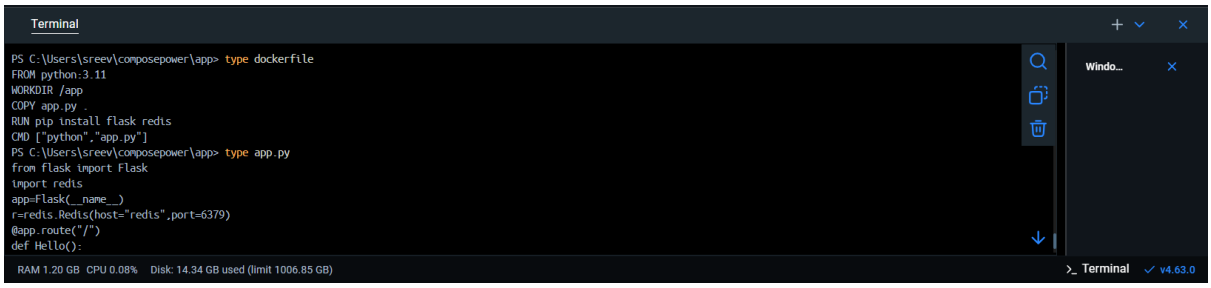
```
Terminal

[+] up 4/4
 ✓ Image composepower-web         Built                    28.0s
 ✓ Network composepower_default   Created                   0.1s
 ✓ Container composepower-redis-1 Created                   0.5s
 ✓ Container composepower-web-1   Created                   0.3s
Attaching to redis-1, web-1
redis-1  | Starting Redis Server
redis-1  | 1:C 11 Mar 2026 08:42:01.949 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
redis-1  | 1:C 11 Mar 2026 08:42:01.950 * Redis version=8.6.1, bits=64, commit=00000000, modified=1, pid=1, just started
redis-1  | 1:C 11 Mar 2026 08:42:01.950 * Configuration loaded
redis-1  | 1:M 11 Mar 2026 08:42:01.951 * monotonic clock: POSIX clock_gettime
redis-1  | 1:M 11 Mar 2026 08:42:01.957 * Running mode=standalone, port=6379.
redis-1  | 1:M 11 Mar 2026 08:42:01.964 * <bf> RedisBloom version 8.6.0 (Git=unknown)
RAM 1.24 GB  CPU 0.08%   Disk: 14.34 GB used (limit 1006.85 GB)
```

```
Terminal

web-1    |  * Debug mode: off
web-1    | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
web-1    |  * Running on all addresses (0.0.0.0)
web-1    |  * Running on http://127.0.0.1:5000
web-1    |  * Running on http://172.23.0.3:5000
web-1    | Press CTRL+C to quit
web-1    | 172.23.0.1 - - [11/Mar/2026 08:42:06] "GET / HTTP/1.1" 200 -
web-1    | 172.23.0.1 - - [11/Mar/2026 08:42:06] "GET /favicon.ico HTTP/1.1" 404 -
web-1    | 172.23.0.1 - - [11/Mar/2026 08:42:09] "GET / HTTP/1.1" 200 -
web-1    | 172.23.0.1 - - [11/Mar/2026 08:42:30] "GET / HTTP/1.1" 200 -
web-1    | 172.23.0.1 - - [11/Mar/2026 08:42:31] "GET / HTTP/1.1" 200 -
web-1    | 172.23.0.1 - - [11/Mar/2026 08:42:36] "GET / HTTP/1.1" 200 -
web-1    | 172.23.0.1 - - [11/Mar/2026 08:43:36] "GET / HTTP/1.1" 200 -
RAM 1.24 GB  CPU 0.08%   Disk: 14.34 GB used (limit 1006.85 GB)
```









4. “Building and Orchestrating a Multi-Container Application using Docker Compose (Node.js + MySQL)”

```
Terminal

PS C:\Users\sreev> mkdir docker-compose-practice1


    Directory: C:\Users\sreev


Mode                 LastWriteTime         Length Name


    Directory: C:\Users\sreev\docker-compose-practice1


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13-03-2026     16:31                app


PS C:\Users\sreev\docker-compose-practice1> cd app
PS C:\Users\sreev\docker-compose-practice1\app> notepad server.js
PS C:\Users\sreev\docker-compose-practice1\app> notepad package.json
PS C:\Users\sreev\docker-compose-practice1\app> ls

RAM 1.65 GB  CPU 0.08%   Disk: 16.41 GB used (limit 1006.85 GB)
```

```
Terminal

yaml: line 1: did not find expected key
PS C:\Users\sreev\docker-compose-practice1> notepad docker-compose.yml
PS C:\Users\sreev\docker-compose-practice1> docker compose up --build
[+] up 1/1
 ✓ Image mysql:latest Pulled                                          8.9s
#1 [internal] load local bake definitions
#1 reading from stdin 569B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
[+] up 1/2rring dockerfile: 2B 0.0s done
 ✓ Image mysql:latest                Pulled                           8.9s
 - Image docker-compose-practice1-app Building                        2.1s
failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory

PS C:\Users\sreev\docker-compose-practice1> cd ./app
PS C:\Users\sreev\docker-compose-practice1\app> ls


    Directory: C:\Users\sreev\docker-compose-practice1\app

RAM 1.39 GB  CPU 1.42%   Disk: 16.41 GB used (limit 1006.85 GB)
```

```
Terminal

PS C:\Users\sreev\docker-compose-practice1\app> ren dockerfile.txt dockerfile
PS C:\Users\sreev\docker-compose-practice1\app> ls


    Directory: C:\Users\sreev\docker-compose-practice1\app


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        13-03-2026     16:55            115 dockerfile
-a----        13-03-2026     16:48            162 package.json
-a----        13-03-2026     16:46            527 server.js


PS C:\Users\sreev\docker-compose-practice1\app> cd ..
PS C:\Users\sreev\docker-compose-practice1> docker compose up --build
#1 [internal] load local bake definitions
#1 reading from stdin 569B done
#1 DONE 0.0s

#2 [internal] load build definition from dockerfile

RAM 1.39 GB  CPU 0.17%   Disk: 16.41 GB used (limit 1006.85 GB)
```

```
Terminal

#12 unpacking to docker.io/library/docker-compose-practice1-app:latest
#12 unpacking to docker.io/library/docker-compose-practice1-app:latest 0.7s done
#12 DONE 1.8s

#13 resolving provenance for metadata file
#13 DONE 0.0s
[+] up 4/4
 ✓ Image docker-compose-practice1-app       Built                     14.4s
 ✓ Network docker-compose-practice1_default Created                    0.1s
 ✓ Container mysql_db                       Created                    0.3s
 ✓ Container node_app                       Created                    0.2s
Attaching to mysql_db, node_app
mysql_db  | 2026-03-13 11:46:59+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.6.0-1.el9 started.
node_app  | Server running on port 3000
node_app  | Database connection failed: Error: connect ECONNREFUSED 172.21.0.2:3306
node_app  |     at TCPConnectWrap.afterConnect [as oncomplete] (node:net:1555:16) {
node_app  |   errno: -111,
node_app  |   code: 'ECONNREFUSED',
node_app  |   syscall: 'connect',
node_app  |   address: '172.21.0.2',
node_app  |   port: 3306,
node_app  |   fatal: true

RAM 1.39 GB  CPU 0.08%   Disk: 16.41 GB used (limit 1006.85 GB)
```

Terminal

```
mysql_db  | 2026-03-13 11:47:14+00:00 [Note] [Entrypoint]: Temporary server stopped
mysql_db  |
mysql_db  | 2026-03-13 11:47:14+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
mysql_db  |
mysql_db  | 2026-03-13T11:47:14.425405Z 0 [System] [MY-015015] [Server] MySQL Server - start.
mysql_db  | 2026-03-13T11:47:14.767535Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 9.6.0) starting as process 1
mysql_db  | 2026-03-13T11:47:14.767555Z 0 [System] [MY-015590] [Server] MySQL Server has access to 12 logical CPUs.
mysql_db  | 2026-03-13T11:47:14.767570Z 0 [System] [MY-015590] [Server] MySQL Server has access to 8163381248 bytes of physical memory.
mysql_db  | 2026-03-13T11:47:14.782581Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysql_db  | 2026-03-13T11:47:15.405205Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysql_db  | 2026-03-13T11:47:15.874673Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysql_db  | 2026-03-13T11:47:15.874784Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this ch
annel.
mysql_db  | 2026-03-13T11:47:15.879755Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to a
ll OS users. Consider choosing a different directory.
mysql_db  | 2026-03-13T11:47:15.921857Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.so
ck
mysql_db  | 2026-03-13T11:47:15.923745Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '9.6.0'  socket: '/var/run/mysqld/mysqld.sock'
 port: 3306  MySQL Community Server - GPL.

View in Docker Desktop   View Config   Enable Watch   Detach
```

RAM 1.39 GB CPU 0.08% Disk: 16.41 GB used (limit 1006.85 GB)

Terminal Update available

localhost:3000

Docker Compose Node + MySQL is running!

Terminal

```
Container mysql_db Stopping
Container mysql_db Killed
Container mysql_db Stopped

PS C:\Users\sreev\docker-compose-practice1> type docker-compose.yml
services:
  app:
    build: ./app
    container_name: node_app
    ports:
     - "3000:3000"
    depends_on:
      - db
  db:
    image: mysql:latest
    container_name: mysql_db
    environment:
      MYSQL_ROOT_PASSWORD: password
      MYSQL_DATABASE: testdb
    ports:
      - "3306:3306"
PS C:\Users\sreev\docker-compose-practice1>
```

RAM 0.98 GB CPU 0.00% Disk: 16.41 GB used (limit 1006.85 GB)

Terminal Update available

Terminal

```
PS C:\Users\sreev\docker-compose-practice1\app> type dockerfile
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node","server.js"]
PS C:\Users\sreev\docker-compose-practice1\app> type server.js
const express=require("express")
const mysql=require("mysql2")
const app=express();
const db=mysql.createConnection({
        host: "db",
        user: "root",
        password: "password",
        database: "testdb"
});
db.connect((err) => {
  if (err) {
    console.log("Database connection failed:", err);
  } else {
```

RAM 0.98 GB CPU 0.00% Disk: 16.41 GB used (limit 1006.85 GB)

Terminal Update available

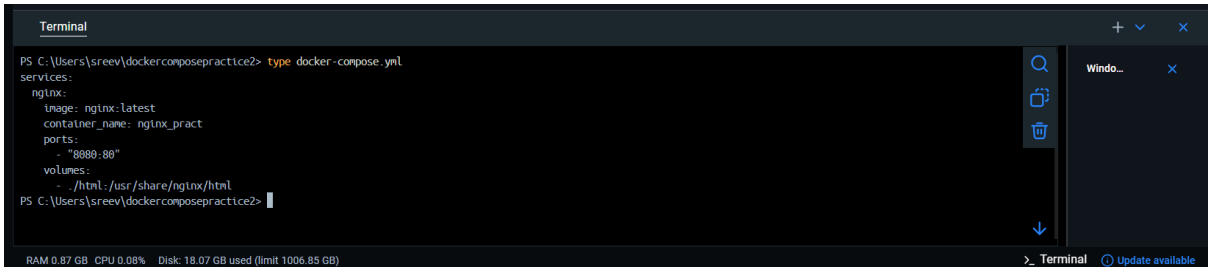
```
Terminal

  console.log("Connected to MySQL!");
  }
});

app.get("/", (req, res) => {
  res.send("Docker Compose Node + MySQL is running!");
});

app.listen(3000, () => {
  console.log("Server running on port 3000");
});
PS C:\Users\sreev\docker-compose-practice1\app> type package.json
{
  "name": "docker-compose-node",
  "version": "1.0.0",
  "main": "server.js",
  "dependencies": {
    "express": "^4.18.2",
    "mysql2": "^3.2.0"
  }
}
PS C:\Users\sreev\docker-compose-practice1\app>

RAM 0.96 GB  CPU 0.00%  Disk: 16.41 GB used (limit 1006.85 GB)
```

5. Containerized Nginx Static Server with Bind Mounts and Port Mapping

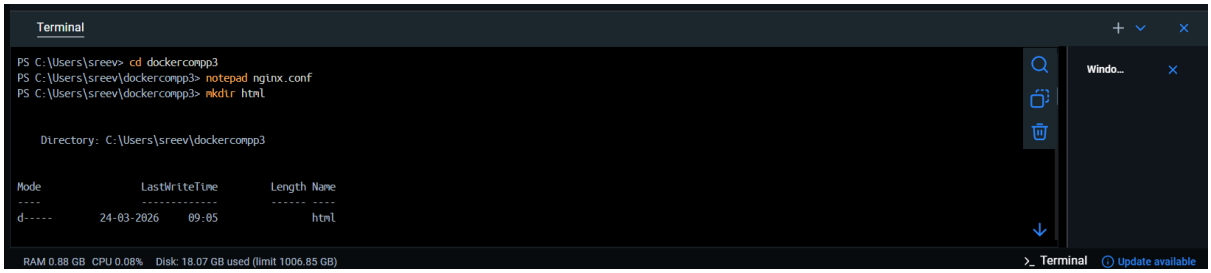
```
Terminal

PS C:\Users\sreev> mkdir dockercomposepractice2


    Directory: C:\Users\sreev


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        24-03-2026     08:46                dockercomposepractice2


RAM 0.88 GB  CPU 4.61%  Disk: 18.07 GB used (limit 1006.85 GB)
```

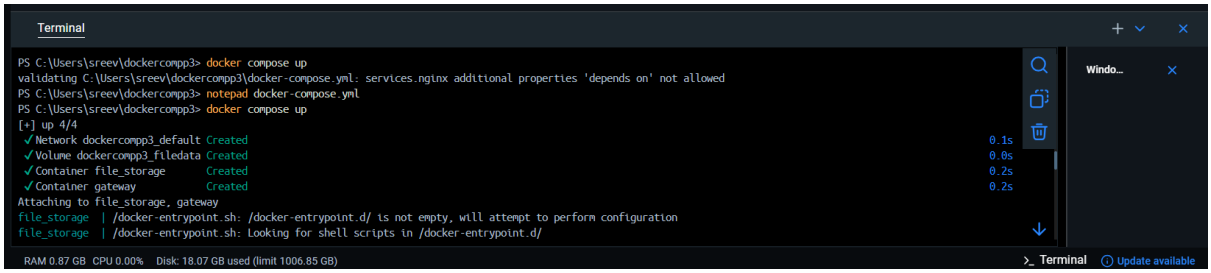
```
Terminal

PS C:\Users\sreev\dockercomposepractice2> notepad docker-compose.yml
PS C:\Users\sreev\dockercomposepractice2> docker compose up
[+] up 2/2
 ✓ Network dockercomposepractice2_default Created                    0.1s
 ✓ Container nginx_pract                  Created                    0.2s
Attaching to nginx_pract
nginx_pract  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
nginx_pract  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
nginx_pract  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
nginx_pract  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
nginx_pract  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
nginx_pract  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh

RAM 0.88 GB  CPU 0.08%  Disk: 18.07 GB used (limit 1006.85 GB)
```

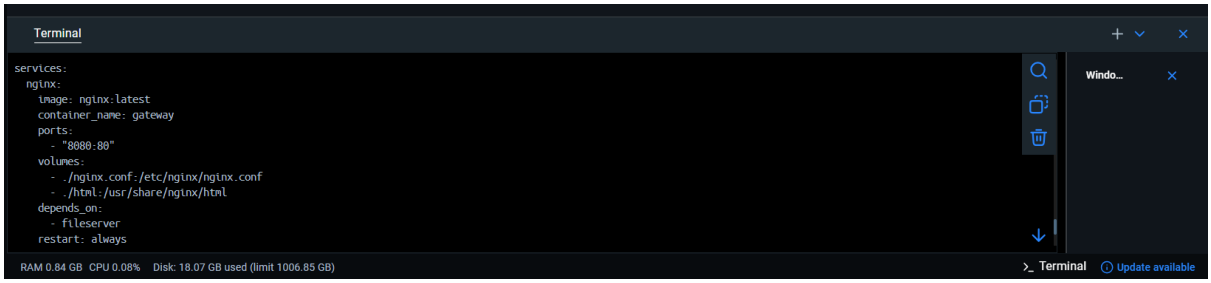
```
Terminal

PS C:\Users\sreev\dockercomposepractice2> mv index.html ./html
PS C:\Users\sreev\dockercomposepractice2> ls


    Directory: C:\Users\sreev\dockercomposepractice2


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        24-03-2026     08:57                html
-a----        24-03-2026     08:53            161 docker-compose.yml

RAM 0.88 GB  CPU 5.34%  Disk: 18.07 GB used (limit 1006.85 GB)
```



6. Nginx Reverse Proxy with Docker Compose and Volume Persistence







```
Terminal
      - ./html:/usr/share/nginx/html
    depends_on:
      - fileserver
    restart: always
  fileserver:
    image: nginx:latest
    container_name: file_storage
    volumes:
      - filedata:/usr/share/nginx/html
volumes:
  filedata:
PS C:\Users\sreev\dockercompp3>
RAM 0.84 GB  CPU 0.00%   Disk: 18.07 GB used (limit 1006.85 GB)
```

7. Implementation of Reverse Proxy Architecture using Nginx and Python Backend with Docker Compose

```
Terminal
PS C:\Users\sreev> mkdir dockercompp4

    Directory: C:\Users\sreev

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        24-03-2026     09:20                dockercompp4

PS C:\Users\sreev> cd dockercompp4
RAM 1.19 GB  CPU 0.42%   Disk: 18.24 GB used (limit 1006.85 GB)
Terminal   Update available
```

```
Terminal
PS C:\Users\sreev> cd dockercompp4
PS C:\Users\sreev\dockercompp4> notepad app.py
PS C:\Users\sreev\dockercompp4> notepad dockerfile
PS C:\Users\sreev\dockercompp4> notepad nginx.conf
PS C:\Users\sreev\dockercompp4> notepad docker-compose.yml
PS C:\Users\sreev\dockercompp4> ren dockerfile.txt dockerfile
PS C:\Users\sreev\dockercompp4> ls

    Directory: C:\Users\sreev\dockercompp4

RAM 1.18 GB  CPU 0.00%   Disk: 18.24 GB used (limit 1006.85 GB)
Terminal   Update available
```

```
Terminal
PS C:\Users\sreev\dockercompp4> docker compose up
#1 [internal] load local bake definitions
#1 reading from stdin 527B done
#1 DONE 0.0s

#2 [internal] load build definition from dockerfile
#2 transferring dockerfile: 112B 0.1s done
#2 DONE 0.2s

#3 [internal] load metadata for docker.io/library/python:3.10-slim
#3 ...
RAM 1.19 GB  CPU 0.17%   Disk: 18.24 GB used (limit 1006.85 GB)
Terminal   Update available
```

```
Terminal
#11 resolving provenance for metadata file
#11 DONE 0.0s
[+] up 4/4
 ✓ Image dockercompp4-backend   Built                    18.0s
 ✓ Network dockercompp4_default Created                  0.1s
 ✓ Container python_backend     Created                  0.4s
 ✓ Container nginx_gateway      Created                  0.3s
Attaching to nginx_gateway, python_backend
nginx_gateway  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
nginx_gateway  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
nginx_gateway  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
RAM 1.18 GB  CPU 0.33%  Disk: 18.24 GB used (limit 1006.85 GB)
```

```
localhost:8080
localhost:8080
Hello from Python Backend
```

```
Terminal
PS C:\Users\sreev\dockercompp4> type docker-compose.yml
services:
  backend:
    build: .
    container_name: python_backend
    restart: always
  nginx:
    image: nginx:latest
    container_name: nginx_gateway
    ports:
      - "8080:80"
    volumes:
RAM 1.16 GB  CPU 0.00%  Disk: 18.24 GB used (limit 1006.85 GB)
```

```
Terminal
      - "8080:80"
    volumes:
      - ./nginx.conf:/etc/nginx/nginx.conf
    depends_on:
      - backend
    restart: always
PS C:\Users\sreev\dockercompp4> type dockerfile
FROM python:3.10-slim
WORKDIR /app
COPY app.py .
CMD ["python","app.py"]
PS C:\Users\sreev\dockercompp4>
RAM 1.16 GB  CPU 0.00%  Disk: 18.24 GB used (limit 1006.85 GB)
```

```
Terminal
from http.server import BaseHTTPRequestHandler, HTTPServer

class Handler(BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header("Content-type", "text/html")
        self.end_headers()
        self.wfile.write(b"Hello from Python Backend ")

server = HTTPServer(("0.0.0.0", 8000), Handler)
print("Server running on port 8000")
server.serve_forever()
RAM 1.16 GB  CPU 0.08%  Disk: 18.24 GB used (limit 1006.85 GB)
```

8. Multi-Container WordPress and MySQL Setup using Docker Compose with Volume Persistence

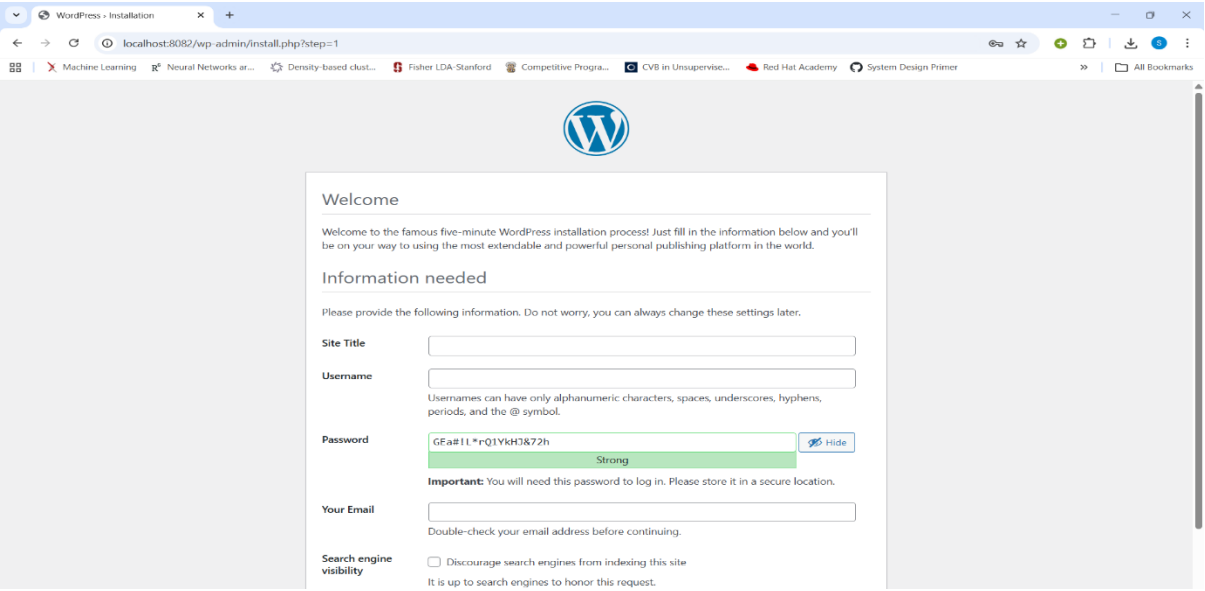
```
Terminal

PS C:\Users\sreev> cd wordpresspract
PS C:\Users\sreev\wordpresspract> notepad docker-compose.yml
PS C:\Users\sreev\wordpresspract> ls


    Directory: C:\Users\sreev\wordpresspract


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        24-03-2026     10:44            688 docker-compose.yml

RAM 3.18 GB  CPU 0.00%   Disk: 20.08 GB used (limit 1006.85 GB)
```

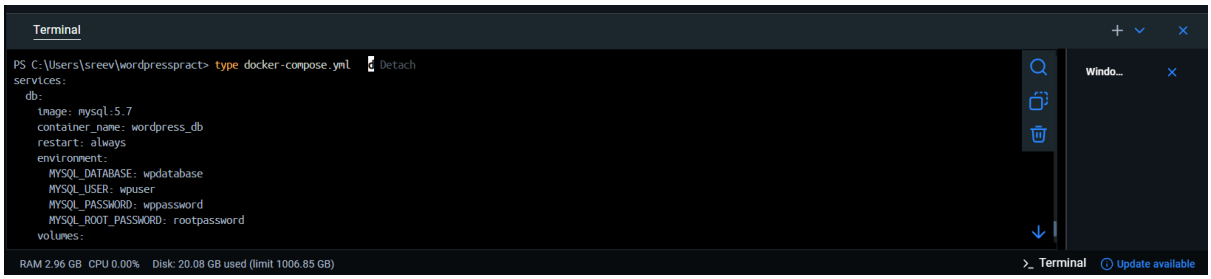
```
Terminal

PS C:\Users\sreev\wordpresspract> docker compose up
[+] up 45/45
 ✓ Image wordpress:latest               Pulled                                   49.9s
 ✓ Image mysql:5.7                      Pulled                                   46.4s
 ✓ Network wordpresspract_default       Created                                   0.2s
 ✓ Volume wordpresspract_wordpress_data Created                                   0.0s
 ✓ Volume wordpresspract_db_data        Created                                   0.0s
 ✓ Container wordpress_db               Created                                   1.0s
 ✓ Container wordpress_app              Created                                   0.3s
Attaching to wordpress_app, wordpress_db
wordpress_db  | 2026-03-24 05:19:17+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
wordpress_db  | 2026-03-24 05:19:18+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'

RAM 3.18 GB  CPU 2.69%   Disk: 20.08 GB used (limit 1006.85 GB)
```

```
Terminal

wordpress_db  | 2026-03-24T05:19:18.981240Z 0 [Warning] Gtid table is not ready to be used. Table 'mysql.gtid_executed' cannot be opened.


wordpress_app | No 'wp-config.php' found in /var/www/html, but 'WORDPRESS_...' variables supplied; copying 'wp-config-docker.php' (WORDPRESS_DB_HOST WORDPRESS_DB_NAME W
ORDPRESS_DB_PASSWORD WORDPRESS_DB_USER)
wordpress_app | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.26.0.3. Set the 'ServerName' directive globally to su
ppress this message
wordpress_app | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.26.0.3. Set the 'ServerName' directive globally to su
ppress this message
wordpress_db  | 2026-03-24T05:19:19.277603Z 0 [Warning] A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
wordpress_db  | 2026-03-24T05:19:19.277680Z 0 [Warning] A deprecated TLS version TLSv1.1 is enabled. Please use TLSv1.2 or higher.
wordpress_db  | 2026-03-24T05:19:19.278916Z 0 [Warning] CA certificate ca.pem is self signed.

RAM 3.18 GB  CPU 0.08%   Disk: 20.08 GB used (limit 1006.85 GB)
```

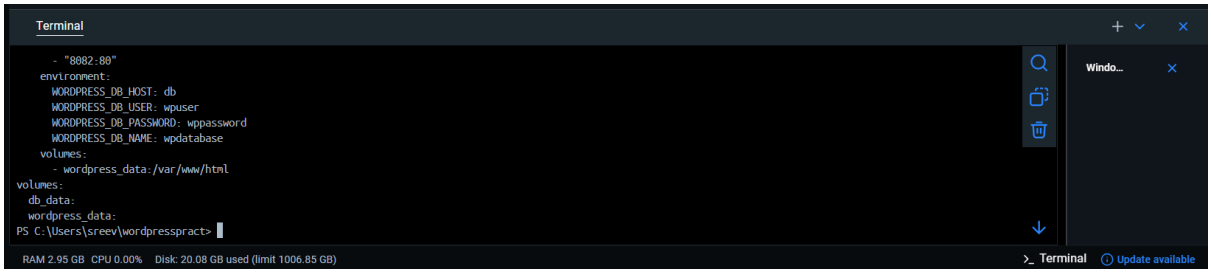
```
Terminal

wordpress_db  | 2026-03-24T05:19:28.937070Z 0 [Note] mysqld: ready for connections.
wordpress_db  | Version: '5.7.44'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server (GPL)
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:42 +0000] "GET / HTTP/1.1" 302 424 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Geck
o) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:42 +0000] "GET /wp-admin/install.php HTTP/1.1" 200 4729 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537
.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:45 +0000] "GET /wp-includes/css/dashicons.min.css?ver=6.9.4 HTTP/1.1" 200 36059 "http://localhost:8082/wp-admin/instal
l.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:45 +0000] "GET /wp-includes/css/buttons.min.css?ver=6.9.4 HTTP/1.1" 200 1808 "http://localhost:8082/wp-admin/install.p
hp" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:45 +0000] "GET /wp-includes/js/jquery/jquery-migrate.min.js?ver=3.4.1 HTTP/1.1" 200 5216 "http://localhost:8082/wp-adm
in/install.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"

RAM 3.18 GB  CPU 0.00%   Disk: 20.08 GB used (limit 1006.85 GB)
```

```
Terminal

admin/install.php?step=1" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:50 +0000] "GET /favicon.ico HTTP/1.1" 302 423 "http://localhost:8082/wp-admin/install.php?step=1" "Mozilla/5.0 (Window
s NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:50 +0000] "GET /wp-includes/js/zxcvbn.min.js HTTP/1.1" 200 400393 "http://localhost:8082/wp-admin/install.php?step=1"
"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 172.26.0.1 - - [24/Mar/2026:05:19:50 +0000] "GET /wp-admin/install.php HTTP/1.1" 200 4729 "http://localhost:8082/wp-admin/install.php?step=1" "Mozilla/5
.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
wordpress_app | 127.0.0.1 - - [24/Mar/2026:05:19:56 +0000] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.66 (Debian) PHP/8.3.30 (internal dummy connection)"
wordpress_app | 127.0.0.1 - - [24/Mar/2026:05:19:57 +0000] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.66 (Debian) PHP/8.3.30 (internal dummy connection)"

View in Docker Desktop   View Config   Enable Watch   Detach

RAM 3.18 GB  CPU 3.96%   Disk: 20.08 GB used (limit 1006.85 GB)
```


WordPress › Installation
localhost:8082/wp-admin/install.php?step=1
Welcome
Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.
Information needed
Please provide the following information. Do not worry, you can always change these settings later.
Site Title
Username
Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.
Password
GEa#!L*rQ1YkH]&72h
Hide
Strong
Important: You will need this password to log in. Please store it in a secure location.
Your Email
Double-check your email address before continuing.
Search engine visibility
Discourage search engines from indexing this site
It is up to search engines to honor this request.


Terminal
PS C:\Users\sreev\wordpresspract> type docker-compose.yml
services:
db:
image: mysql:5.7
container_name: wordpress_db
restart: always
environment:
MYSQL_DATABASE: wpdatabase
MYSQL_USER: wpuser
MYSQL_PASSWORD: wppassword
MYSQL_ROOT_PASSWORD: rootpassword
volumes:
RAM 2.96 GB CPU 0.00% Disk: 20.08 GB used (limit 1006.85 GB)
Terminal
Update available

Terminal
- db_data:/var/lib/mysql
wordpress:
image: wordpress:latest
container_name: wordpress_app
restart: always
depends_on:
- db
ports:
- "8082:80"
environment:
WORDPRESS_DB_HOST: db
WORDPRESS_DB_USER: wpuser
RAM 2.95 GB CPU 0.00% Disk: 20.08 GB used (limit 1006.85 GB)
Terminal
Update available


Terminal
- "8082:80"
environment:
WORDPRESS_DB_HOST: db
WORDPRESS_DB_USER: wpuser
WORDPRESS_DB_PASSWORD: wppassword
WORDPRESS_DB_NAME: wpdatabase
volumes:
- wordpress_data:/var/www/html
volumes:
db_data:
wordpress_data:
PS C:\Users\sreev\wordpresspract>
RAM 2.95 GB CPU 0.00% Disk: 20.08 GB used (limit 1006.85 GB)
Terminal
Update available

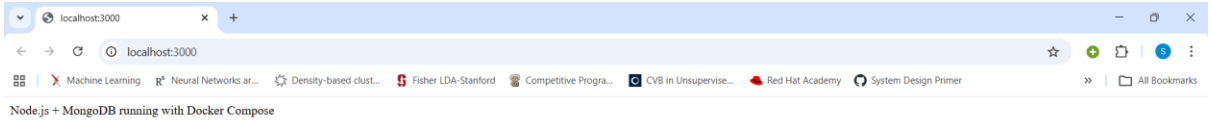
9. Containerized Node.js Backend with MongoDB using Docker Compose, Dockerfile, and Named Volumes

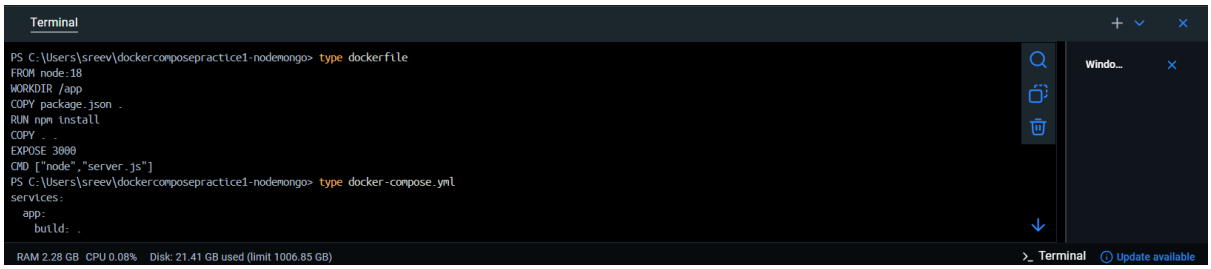
```
Terminal

Directory: C:\Users\sreev\dockercomposepractice1-nodemongo

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        24-03-2026     11:15            335 docker-compose.yml
-a----        24-03-2026     11:10            113 dockerfile
-a----        24-03-2026     11:07            141 package.json
-a----        24-03-2026     11:06            483 server.js

PS C:\Users\sreev\dockercomposepractice1-nodemongo> docker compose up --build

RAM 2.38 GB  CPU 0.00%   Disk: 21.41 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

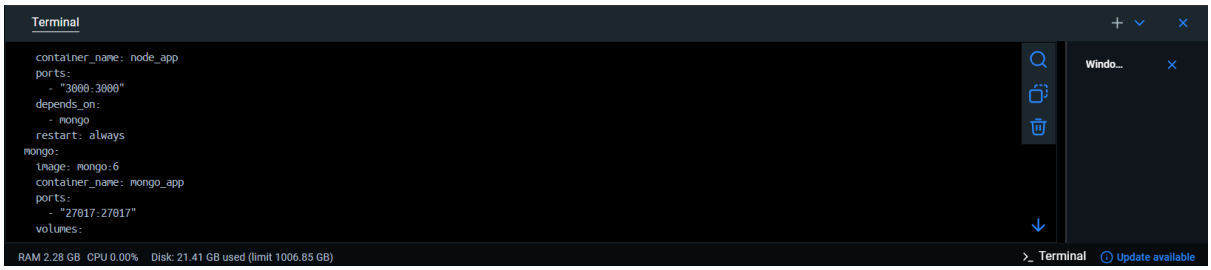
```
Terminal

PS C:\Users\sreev\dockercomposepractice1-nodemongo> notepad docker-compose.yml
PS C:\Users\sreev\dockercomposepractice1-nodemongo> docker compose up --build
[+] up 11/11
 ✓ Image mongo:6 Pulled                                                    44.8s
#1 [internal] load local bake definitions
#1 reading from stdin 591B done
#1 DONE 0.0s

#2 [internal] load build definition from dockerfile
#2 transferring dockerfile: 150B 0.1s done
#2 DONE 0.2s

RAM 2.38 GB  CPU 0.17%   Disk: 21.41 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

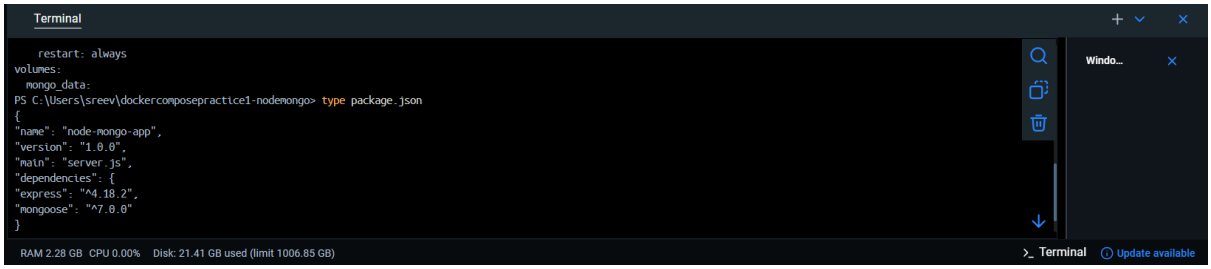
```
Terminal

[+] up 16/16g provenance for metadata file
 ✓ Image mongo:6                                    Pulled                44.8s
 ✓ Image dockercomposepractice1-nodemongo-app       Built                 8.1s
 ✓ Network dockercomposepractice1-nodemongo_default Created               0.1s
 ✓ Volume dockercomposepractice1-nodemongo_mongo_data Created             0.0s
 ✓ Container mongo_app                              Created               0.5s
 ✓ Container node_app                               Created               0.3s
Attaching to mongo_app, node_app
mongo_app  | {"t":{"$date":"2026-03-24T05:46:42.790+00:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"-","msg":"Initialized wire specification","attr":{"spec":{"inc
omingExternalClient":{"minWireVersion":0,"maxWireVersion":17},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":17},"outgoing":{"minWireVersion":6,"maxWireVe
rsion":17},"isInternalClient":true}}}

RAM 2.38 GB  CPU 0.25%   Disk: 21.41 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```

```
Terminal

ilding indexes":"1 ms","Drop abandoned idents and get back indexes that need to be rebuilt or builds that need to be restarted":"0 ms","Drop abandoned idents and get bac
k indexes that need to be rebuilt or builds that need to be restarted":"0 ms","Rebuild indexes for collections":"0 ms","Rebuild indexes for collections":"0 ms","Build us
er and roles graph":"0 ms","Set up the background thread pool responsible for waiting for opTimes to be majority committed":"0 ms","Start up the replication coordinator"
:"0 ms","Start transport layer":"1 ms","_initAndListen total elapsed time":"524 ms"}}}}
mongo_app  | {"t":{"$date":"2026-03-24T05:46:43.409+00:00"},"s":"I",  "c":"INDEX",    "id":20345,   "ctx":"LogicalSessionCacheRefresh","msg":"Index build: done building"
,"attr":{"buildUUID":null,"collectionUUID":{"uuid":{"$uuid":"52f4a071-08b7-41f6-b1d3-c5da6f2cd06c"}},"namespace":"config.system.sessions","index":"_id_","ident":"index-5
-1741854607905562808","collectionIdent":"collection-4-1741854607905562808","commitTimestamp":null}}
mongo_app  | {"t":{"$date":"2026-03-24T05:46:43.409+00:00"},"s":"I",  "c":"INDEX",    "id":20345,   "ctx":"LogicalSessionCacheRefresh","msg":"Index build: done building"
,"attr":{"buildUUID":null,"collectionUUID":{"uuid":{"$uuid":"52f4a071-08b7-41f6-b1d3-c5da6f2cd06c"}},"namespace":"config.system.sessions","index":"lsidTTLIndex","ident":
"index-6-1741854607905562808","collectionIdent":"collection-4-1741854607905562808","commitTimestamp":null}}
node_app   | Server running on port 3000

RAM 2.38 GB  CPU 0.08%   Disk: 21.41 GB used (limit 1006.85 GB)   >_ Terminal   Update available
```


localhost:3000
localhost:3000
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Node.js + MongoDB running with Docker Compose


Terminal
PS C:\Users\sreev\dockercomposepractice1-nodemongo> type dockerfile
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node","server.js"]
PS C:\Users\sreev\dockercomposepractice1-nodemongo> type docker-compose.yml
services:
app:
build: .
Windo...
RAM 2.28 GB CPU 0.08% Disk: 21.41 GB used (limit 1006.85 GB)
Terminal
Update available


Terminal
container_name: node_app
ports:
- "3000:3000"
depends_on:
- mongo
restart: always
mongo:
image: mongo:6
container_name: mongo_app
ports:
- "27017:27017"
volumes:
Windo...
RAM 2.28 GB CPU 0.00% Disk: 21.41 GB used (limit 1006.85 GB)
Terminal
Update available


Terminal
restart: always
volumes:
mongo_data:
PS C:\Users\sreev\dockercomposepractice1-nodemongo> type package.json
{
"name": "node-mongo-app",
"version": "1.0.0",
"main": "server.js",
"dependencies": {
"express": "^4.18.2",
"mongoose": "^7.0.0"
}
Windo...
RAM 2.28 GB CPU 0.00% Disk: 21.41 GB used (limit 1006.85 GB)
Terminal
Update available